

	decrease.	
3(a)	The first law of thermodynamics states that the increase in internal energy of a system is the sum of the heat supplied to the system and the work done on the system.	[1] correct definition
3(b)(i)	$\Delta U = Q + W$ For adiabatic process, $Q = 0$ $\Delta U = W = -500 \text{ J}$	[1] correct answer
3(b)(ii)	Since internal energy decreases, <u>temperature of gas decreases.</u>	[1] correct answer
3(c)(i)	$\Delta U = Q + W$ For constant volume process, $W = 0$ $Q = \Delta U = \frac{3}{2} nR \Delta T$ $= \frac{3}{2} (2.5)(8.31)(500 - 300)$ $= 6232.5$ $= 6230 \text{ J}$ $\text{molar heat capacity} = \frac{6232.5}{(2.5)(200)}$ $= 12.465$ $= 12.5 \text{ J mol}^{-1} \text{ K}^{-1}$	[1] for correct equation and substitution [1] for correct Q [1] for correct molar heat capacity

No.	Solution	Remarks
3(c)(ii)	For the constant pressure process, in addition to the increase in internal energy, work is done by the gas hence more heat is supplied for the same change in temperature. This will result in a larger molar heat capacity.	[1] for work done [1] for more heat supplied
4(a)	V V	